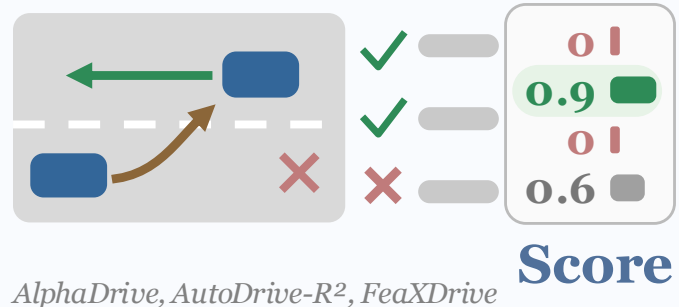


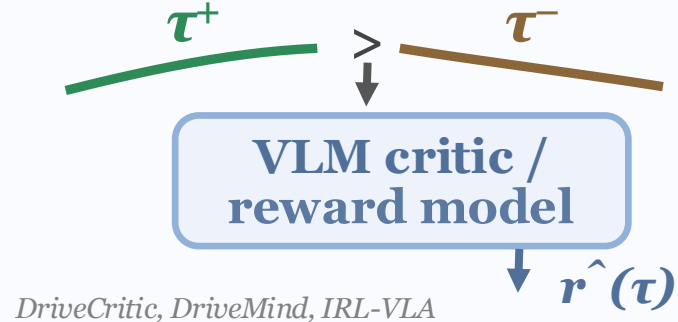
Reward Design as Supervision Construction

— Sec. 5.1

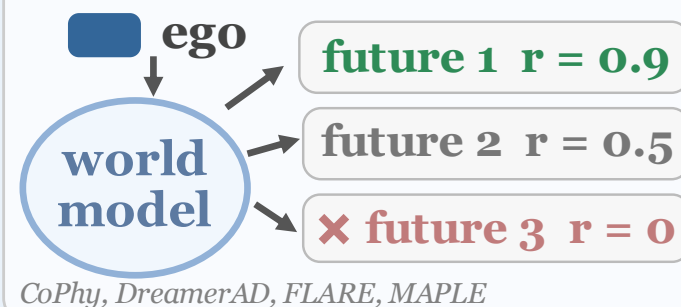
(1) Rule-based / Metric



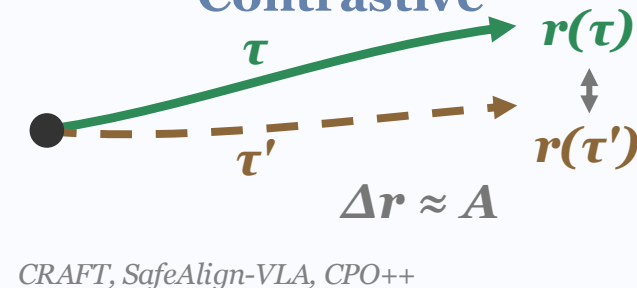
(2) Learned Reward Models



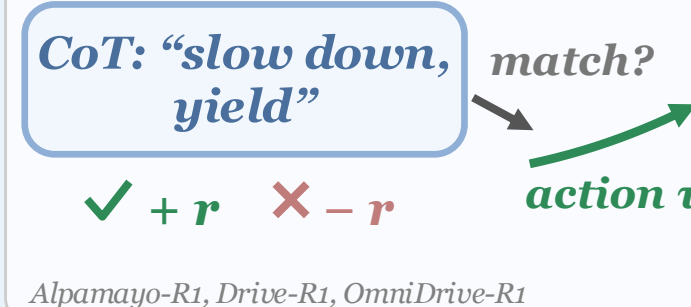
(3) World-Model-Coupled



(4) Counterfactual / Contrastive



(5) Reasoning Consistency



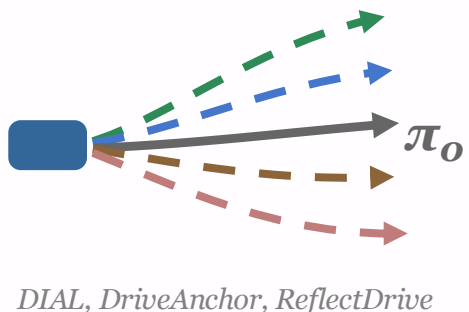
rollouts τ

rewards $r(\tau)$

Rollout Collection & Informative Data

— Sec. 5.2

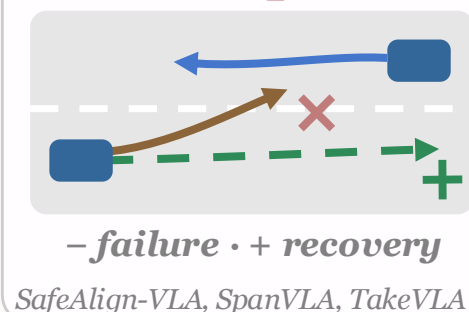
(1) Enhancing Behavior



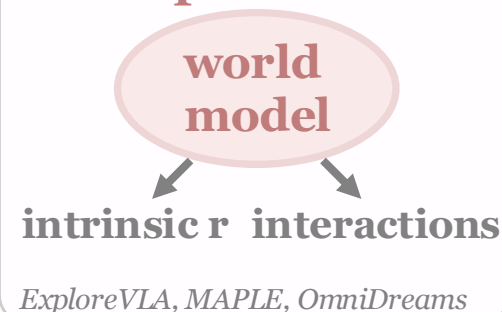
(2) Group Diversity



(3) Negative-Sample



(4) World-Model Exploration

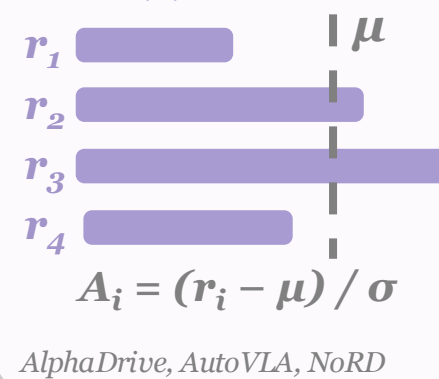


updated policy π

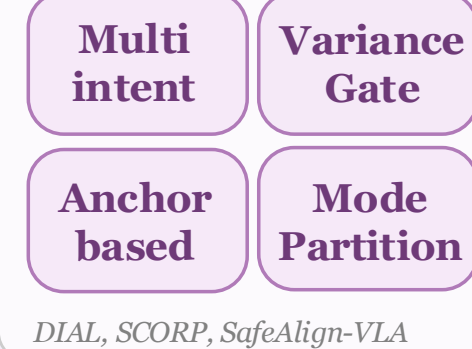
Policy Optimization

— Sec. 5.3

(1) GRPO



(2) GRPO Variants



(3) Value Based PO

